

MEMS 3430 Final Project: Ice Rink Cooling System

Maxwell Mitchell

Peter Nau

Brendan Rubel

Joe Gillett

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Executive Summary

This report outlines the design and optimization of a commercial refrigeration system built to maintain a regulation-sized indoor ice hockey rink in St. Louis, MO. Maintaining an indoor sheet of ice during a hot Midwestern summer requires significant energy. Therefore, the primary goal of this project was to design a system that is not only capable of keeping the ice frozen on the hottest days of the year, but also operates with high energy efficiency and a low environmental footprint.

To prioritize public safety, the facility uses an indirect cooling method. Instead of pumping hazardous chemical refrigerants directly under the ice where skaters are present, the system chills a safe, antifreeze-like fluid (propylene glycol). This cold fluid is continuously pumped through pipes below the rink floor to absorb heat from the ice. That heat is then carried back to an isolated mechanical room, transferred to a modern, environmentally friendly refrigeration cycle, and ultimately exhausted to the warm outdoor air.

Our team analyzed the complete life-cycle of this process, calculating the exact equipment sizes and power requirements needed to beat the summer heat. Furthermore, we optimized the system by adjusting pressure levels of the refrigerant to find the best balance between upfront equipment costs and long-term electricity savings. The final design demonstrates that by utilizing modern refrigerants and optimized operating temperatures, a St. Louis ice rink can maintain safe, professional-grade ice while successfully minimizing both its carbon footprint and its monthly utility bills.

1 Introduction and Literature Review

Motivation

Ice rink cooling systems represent one of the most energy-intensive commercial HVAC applications due to the massive, continuous cooling load required to maintain a frozen surface within an occupied building. The design of a thermal cooling system for an ice rink was chosen because it requires a complex balancing of thermodynamic efficiency, environmental sustainability, and strict safety constraints. With the ongoing phase-out of high-global-warming-potential (GWP) refrigerants like R-22, optimizing these thermal systems is critical to reducing both operational costs and greenhouse gas emissions for ice rinks. [1].

State of the Art and Existing Systems

Modern ice rink refrigeration systems are broadly categorized into direct and indirect systems. Direct systems pump the primary refrigerant directly through the rink floor, but these have largely

been abandoned due to the severe safety hazards associated with large refrigerant leaks [2]. Consequently, the current state-of-the-art approach utilizes an indirect system. In this setup, a primary refrigeration plant cools a secondary heat transfer fluid, typically a glycol or calcium chloride brine, which is then circulated beneath the ice slab [2]. This separation ensures that the primary refrigerant is safely isolated within the mechanical room.

Within indirect systems, the choice of the primary refrigeration cycle dictates the system's overall efficiency and environmental footprint. Currently, three approaches dominate the industry:

- **Ammonia (R-717) Systems:** Ammonia has historically been the industry standard for large arenas due to its exceptional thermodynamic efficiency and zero GWP. However, because ammonia is highly toxic, its use requires strict safety regulations, extensive mechanical room isolation, and dedicated on-site engineering staff to ensure public safety [3].
- **Transcritical CO₂ (R-744) Systems:** Transcritical CO₂ cycles represent the state-of-the-art in community rinks and National Hockey League arenas [1]. CO₂ is non-toxic, non-flammable, and has a baseline GWP of 1. These systems operate at much higher pressures than traditional cycles, but offer superior heat recovery capabilities; the rejected heat from the gas cooler can be captured to provide space heating, dehumidification, and hot water for the entire facility, drastically improving overall building efficiency [1].
- **Synthetic HFO Refrigerant Systems:** For facilities where the high operating pressures of CO₂ or the toxicity of ammonia are prohibitive, modern hydrofluoroolefin (HFO) blends are increasingly utilized [3]. These systems operate at standard pressure ranges, utilize familiar compressor technology, and provide a balance of low toxicity, low flammability, and significantly lower GWP than legacy Freon systems [3].

By studying these existing frameworks, our project aims to model a highly efficient indirect cooling cycle that successfully manages the thermal loads of a regulation-sized rink while adhering to modern sustainability and safety standards.

2 Boundary Conditions and Assumptions

Boundary Conditions

The physical and environmental constraints for the ice rink cooling system, focusing on peak summer conditions in St. Louis, MO, are defined as follows:

- **Location and Climate:** The facility is located in St. Louis, Missouri, classified as ASHRAE Climate Zone 4A [4].

- **Peak Outdoor Temperatures:** The system must reject heat to the environment at the 1% summer design dry-bulb temperature of 91°F (32.8°C) and a coincident wet-bulb temperature of 75°F (23.9°C) [4].
- **Rink Dimensions:** The ice surface follows standard NHL regulation dimensions of 200 ft × 85 ft, creating a refrigerated surface area of approximately 17,000 sq ft (1,580 m²) [5].
- **Target Ice Temperature:** The operational requirement for the ice surface temperature is fixed at −5.5°C to support standard ice hockey activities [6].
- **Indoor Ambient Conditions:** The indoor air temperature directly above the rink is maintained at 12.8°C [6].
- **Compressor Efficiency:** The primary refrigerant compressor is modeled with an isentropic efficiency of $\eta_s = 80\%$, representing realistic performance for a modern commercial screw compressor operating under typical cooling loads [7].
- **Condenser Approach Temperature:** An air-cooled condenser is assumed with a design approach temperature of 8.3°C above the ambient outdoor dry-bulb temperature, yielding a design condensing temperature of 41.1°C (calculated from the 32.8°C outdoor temperature +8.3°C) [8].

System-Level Assumptions

To perform the thermodynamic cycle analysis and optimize the refrigeration system, the following standard assumptions are applied:

- **System Model:** Steady-state operation with negligible kinetic and potential energy changes ($\Delta KE \approx 0, \Delta PE \approx 0$) is assumed [9].
- **Heat Losses & Pressure Drops:** All connecting piping is assumed to be perfectly insulated ($\dot{Q}_{loss} = 0$) and isobaric outside of the compressor and expansion valve ($\Delta P = 0$). These are standard design simplifications used to isolate the thermodynamic performance [9].
- **Expansion Process:** The flow through the expansion valve is modeled as an isenthalpic throttling process ($h_{in} = h_{out}$), representing the universally accepted theoretical model for standard refrigeration expansion devices [9].
- **Primary Refrigerant:** The primary refrigeration cycle is modeled using R-513A, a modern hydrofluoroolefin (HFO) blend. This selection aligns with the literature review’s focus on low-GWP alternatives while maintaining standard subcritical operating pressures, avoiding

the complex hardware requirements of transcritical CO₂ [3]. However, during modeling, R-134A was used, as the properties are readily available in CoolProp and X-steam, and are very similar to R-513A as R-513A is composed of 44% R-134A. [10].

- **Secondary Coolant:** A secondary loop utilizing a 30% by mass propylene glycol-water mixture is assumed. This is standard industry practice for indoor rinks to prevent potentially toxic primary refrigerants from entering the occupied rink space [6].
- **Floor Temperature Approach:** To maintain the -5.5°C ice surface against the thermal resistance of the concrete and piping network, the secondary coolant supply temperature is assumed to be -11.1°C (a 5.6°C differential), aligning with standard heat transfer guidelines for ice rink floors [6].
- **Heat exchanger cross section based on DHW 3** To simplify the heat exchanger analysis, baffle spacings and pitch lengths, as well as approximations for shell size were taken from DHW 3. The number of tubes within the exchanger was modified to adjust for a reasonable length of exchanger, but other properties remained constant.

3 System Layout and Governing Equations

Diagrams

Figure 1 presents TS diagrams for the brine and refrigerant sides of the system. For the brine specifically, the cycle can be modeled on a single isobar, as the liquid is cycled from below the ice to within the heat exchanger, at essentially constant pressure. It also presents a TX diagram within the heat exchanger, with the temp of the refrigerant constant due to phase change.

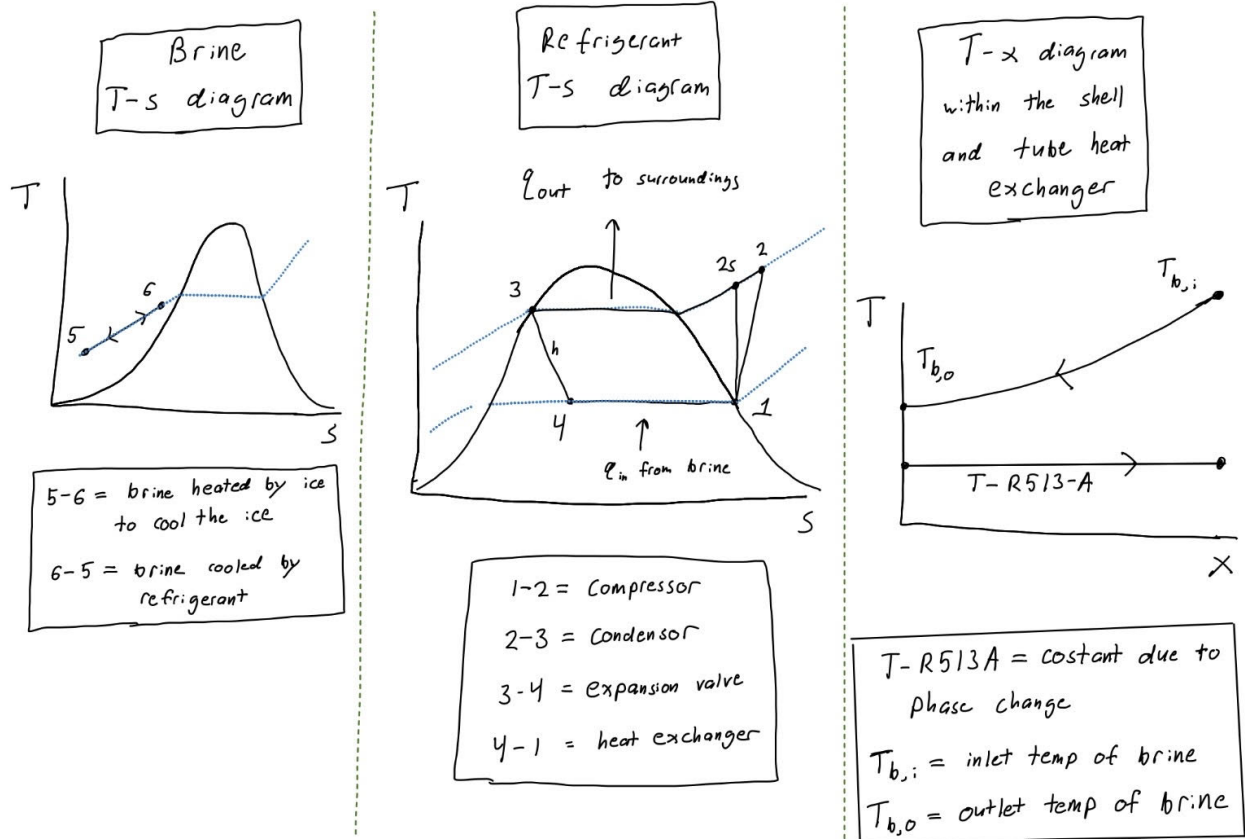


Figure 1: TS and TX Diagrams of system components

Governing Equations

The solution to this system draws heavily on the work performed in design homework assignments 2 and 3. To begin, the total heat gained per second by the surface in watts $\dot{Q}_{cool}$ is found in Eq. 1 as:

$$\dot{Q}_{cool} = \dot{Q}_{conv} + \dot{Q}_{rad} + \dot{Q}_{diff} + \dot{Q}_{resurface} \quad (1)$$

where $\dot{Q}_{conv}$ is the heat gained due to convection, $\dot{Q}_{rad}$ is the heat gained due to incident radiation, $\dot{Q}_{diff}$ is the heat gained due to water condensation on the surface of the ice, and $\dot{Q}_{resurface}$ is the heat gained during the ice resurfacing process. All other heat sources are assumed to be negligible.

The total heat gained due to convection depends on the convection coefficient of the system h_{conv} as well as the area of the ice in m^2 A_{ice} and the temperature difference between the area and the ice in Celsius $T_{air} - T_{ice}$. Eq. 2 demonstrates this relationship:

$$\dot{Q}_{conv} = h_{conv}A_{ice}(T_{air} - T_{ice}) \quad (2)$$

The total heat gained due to radiation depends on the emissivity of the surroundings ε , the Boltzmann constant σ , and the temperatures T_{air} and T_{ice} . Eq. 3 demonstrates this relationship:

$$\dot{Q}_{rad} = \varepsilon\sigma A_{ice}(T_{ceiling}^4 - T_{ice}^4) \quad (3)$$

As vapor in the surrounding environment condensates on the rink, heat is transferred into the rink following Eq. 4. This depends on A_{ice} and the difference between T_{air} and T_{ice} , as well as h_{diff} , the coefficient of diffusion. h_{diff} is given by Eq. 5:

$$\dot{Q}_{diff} = h_{diff}A_{ice}(T_{air} - T_{ice}) \quad (4)$$

$$h_{diff} = 1740(3.41 + 3.55v_{air})\left(\frac{\phi P_{air} - P_{ice}}{T_{air} - T_{ice}}\right) \quad (5)$$

where v_{air} is the latent air velocity in m/s, ϕ is the relative humidity of the air above the rink, and $P_{air} - P_{ice}$ is the difference in vapor saturation pressures in the air and at the ice's surface.

$\dot{Q}_{surface}$ provides the heat accumulated by the rink during the resurfacing process where water is poured over the rink and allowed to freeze, shown in Eq. 6. It depends on V_f , the water volume in m^3 , the water temperature T_{water} , and the ice temperature T_{ice} .

$$\dot{Q}_{resurface} = 1000V_f(4.2(T_{water} - 0) + 334 + 2(0 - T_{ice})) \quad (6)$$

It is assumed that 100% of the heat absorbed by the rink is transferred from the ice to the brine cooling tubes below. The thermodynamic cycle analysis largely mirrored work performed during design homework 2[11]. The core of this work relies on Eq. 7, where the section head transfer in watts $\dot{Q}_{section}$ is given as:

$$\dot{Q}_{section} = \dot{m}(h_B - h_A) \quad (7)$$

where $\dot{m}$ is the mass flow rate of the refrigerant fluid in kg/s, while h_A and h_B are the inlet and outlet enthalpies respectively. The enthalpies were found using the CoolProp module in python. The coefficient of performance (COP) was calculated using Eq. 8:

$$COP = \frac{\dot{Q}_{evap}}{\dot{W}_{comp}} \quad (8)$$

where $\dot{Q}_{evap}$ is the heat absorbed by the refrigerant in the evaporator in watts and $\dot{W}_{comp}$ is the

power supplied to the compressor in watts.

To determine the size of the heat exchanger, an $\varepsilon - NTU$ method was used. For the brine side of the exchanger (within the tubes), the same equations were used as in DHW 3[12] (included in the appendix), with the only exception being equation 9, as the Reynolds number for flow within the pipe is below the 2300 turbulent cutoff.

$$\bar{N}U = 3.66 \quad (9)$$

However, as there is phase change on the shell side of the exchanger (in the refrigerant), a deviation is required from the typical equations, as Cr approaches 0 due to the phase change. As this is the case, equation 12 was utilized, and to solve for equations 10 and 11, C_{min} was evaluated from the Cp and $\dot{m}$ values of the brine. q is the heat transferred in W, C_{min} is the product of mass flow rate and c_p for the brine, and $T_{h,i} - T_{c,i}$ represents the difference between the brine and refrigerant temperatures respectively. ε is the effectiveness, and NTU is a dimensionless parameter.

$$q_{max} = C_{min}(T_{h,i} - T_{c,i}) \quad (10)$$

$$\varepsilon = q/q_{max} \quad (11)$$

$$\varepsilon = 1 - \exp(-NTU) \quad (12)$$

4 Thermodynamic Cycle Analysis and Cycle Equations

State Determination

Properties were determined using thermodynamic property tables for R-134A, with the knowledge that there are 4 unique states with $P_{low} = 155.1$ kPa and $P_{high} = 1049.6$ kPa. The compressor has an isentropic efficiency of $\eta_s = 80\%$. This is based on [13] and [14].

- **State 1 - Evaporator Exit:** Saturation vapor is $T_1 = -16.7^\circ\text{C}$.

$$- h_1 = 236.4 \text{ kJ/kg}, s_1 = 0.938 \text{ kJ/(kg} \cdot \text{K)}, P_1 = 155.1 \text{ kPa}$$

- **State 2 - Compressor Exit:** Superheated gas at $P_2 = 1049.6$ kPa.

$$- h_{2s} = 278.2 \text{ kJ/kg}, h_2 = h_1 + \frac{h_{2s} - h_1}{\eta_s} = 236.4 + \frac{278.2 - 236.4}{0.80} = 288.7 \text{ kJ/kg}, T_2 = 54.9^\circ\text{C}$$

- **State 3 - Condenser Exit:** Saturated liquid at $T_3 = 41.1^\circ\text{C}$.

$$- T_3 = 41.1^\circ\text{C}, h_3 = 158.17 \text{ kJ/kg}, P_3 = 1049.6 \text{ kPa}$$

- **State 4 - Evaporator Inlet:** Contains a liquid-vapor mixture.

$$- T_4 = -16.7^\circ\text{C}, h_4 = h_3 = 158.17 \text{ kJ/kg}, x_4 = \frac{h_4 - h_f}{h_g - h_f} = \frac{158.17 - 78.4}{288.5 - 78.4} = 0.38 = 38\%$$

Mass Flow and Component Loads

By using the enthalpies found the the system's states and the cooling load of $Q_{evap} = 279.8 \text{ kW}$, the following following properties were found:

$$Q_{evap} = \dot{m}_{refrigerant}(h_1 - h_4) = 2.147(288.52 - 158.17) = 279.8 \text{ kW} \quad (13)$$

$$\dot{m}_{refrigerant} = \frac{Q_{evap}}{h_1 - h_4} = \frac{279.8}{288.52 - 158.17} = 2.147 \text{ kg/s} \quad (14)$$

$$W_{comp} = \dot{m}_{refrigerant}(h_2 - h_1) = 2.147(334.72 - 288.52) = 99.2 \text{ kW} \quad (15)$$

$$Q_{condenser} = Q_{evap} + W_{comp} = 279.8 + 99.2 = 379 \text{ kW} \quad (16)$$

$$COP = \frac{Q_{evap}}{W_{comp}} = \frac{279.8}{99.2} \approx 2.82 \quad (17)$$

Brine Loop Analysis

In order to transfer heat from the rink to the primary refrigeration cycle, the study used a 30% propylene glycol solution. In order to find the mass flow rate of the brine, approach temperatures were used with the cooling load for calculations.

- Inlet Temperature ($T_{h,i}$): -5.5°C (Returning from the rink)
- Outlet Temperature ($T_{h,o}$): -11.1°C (Supplying to the rink)
- Specific Heat ($C_{p,brine}$): $\approx 3.8 \text{ kJ/(kg} \cdot \text{K)}$ (C_p for 30% glycol at -8°C)

$$\dot{m}_{brine} = \frac{Q_{evap}}{C_{p,brine}(T_{h,i} - T_{h,o})} = \frac{279.8}{3.8(-5.5 - (-11.1))} \approx 12.07 \text{ kg/s} \quad (18)$$

5 Analysis and Optimization

In order to determine the thermodynamic efficiency of the ice rink cooling system, the results depend on the operating conditions. The system was optimized by focusing the the primary cycle's evaporator pressure and its saturation temperature. The analysis will aim to find the a preferred balance between maintaining and efficient Coefficient of Performance while still obtaining an efficient thermal gradient used for the brine loop. The evaporator pressure will alter the evaporating temperature base on the saturation curve.

Trade-offs

The system's COP was analyzed over a varying range of evaporating temperatures and pressures. The boundary condition for the secondary brine loop supply was set at -11.1°C to maintain -5.5°C for the ice surface.

- **High P_{low} :** When increasing the suction pressure of the compressor, the pressure ratio is minimized. This results in lower compressor work and an increased system COP. When raising the pressure, the refrigerant's boiling temperature increases. If the temperature gets too close to the brine supply temperature, the system will need an incredibly large heat exchanger to transfer the heat.
- **Low P_{low} :** Dropping the suction pressure will lower the evaporating temperature, which will cause the compressor to require a large pressure lift to reach the condenser pressure. As a result the electrical power consumption will raise and the COP will lower.

Optimization results and selected Design

Figure 2 represents the visualization of the system's thermodynamic trade-offs. Since the system uses a boiling refrigerant, the plots rely upon the R-134a saturation curve. The top plot represents the evaporator temperature range and the bottom represents the evaporator pressure range.

- **Efficiency Gain:** Both plots have show a line that represents an increase in the system's COP as the evaporating temperature and pressure increase (T_1 and $P_1 = P_4$). Therefore, as both x values increase, the more efficient the system becomes. A higher evaporator pressure results in less required lift for the compressor and less required electricity.
- **Selected design point:** The red dot represents the selected operating state for the primary cycle at -16.7°C and 1049.6 kPa. Because of the design constraints of the problem, T_1 cannot be increased, as the refrigerant temperature would move closer to the brine supply

requirement of -11.1°C and would decrease the thermal driving force. However, as can be seen in the graphs, T_1 also cannot decrease without decreasing COP, which confirms the assertion that -16.7 is the correct temperature for state 1. The same is true for the evaporator pressure, as decreasing this pressure would lead to a lower COP, while increasing would also lead to less thermodynamic driving force as state 1 is fixed at the saturated liquid line, so increasing pressure also increases temperature.

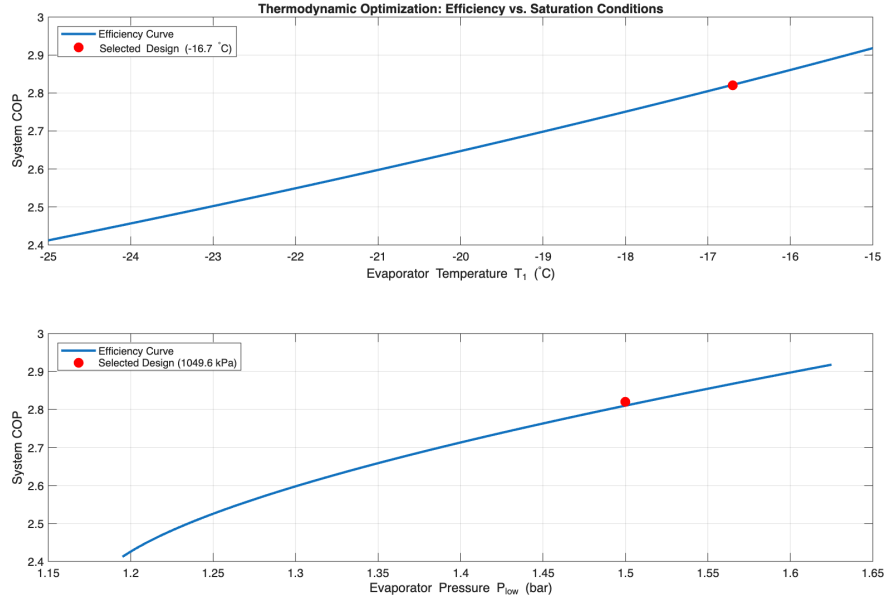


Figure 2: System COP compared to Evaporator Temp (top) and Evaporator Pressure (bottom). Created with the Assistance of AI.

In order to ensure effective heat transfer while lowering the facility's carbon footprint, an evaporating temperature of -16.7°C and a pressure of 1049.6 kPa was determined, and now confirmed through these graphs as optimal. At these points, the system obtains a COP of 2.82. These points in the system allow for high system efficiency while maintaining a safe approach temperature when compared to the brine. Additionally, through the use of the equation presented above to apply the eta-NTU method, and those included below in the Appendix, Python code was created to solve for the length of heat exchanger, which was determined to be 15.184 meters in length.

6 Financial and Environmental Impact

Financial Impact and Operational Expenditures

To find the financial reasonability for the ice rink cooling system, the operational expenditures to support the cooling load was calculated. The ice rink requires a peak thermal load of 316 kW.

The R-513A cycle in Section 4 will be treated as the primary cycle. The R-513A cycle was modeled as an R-134a, and the system achieved a Coefficient of Performance (COP) of 2.82. The compressor requires a peak electrical input of 99.2 kW. Since the ice rink compressor runs continuously during the peak summer, the average St. Louis commercial electricity rate was calculated as \$0.11 per kWh [15]. The assumed average daily electrical cost peak at \$261.89.

Optimizing the evaporator saturation pressure and temperature, the ice rink requires a higher upfront capital expenditure. Higher evaporating temperature limits the temperature approach of the brine, which needs a larger heat exchanger to achieve the same cooling capacity. An increased upfront capital expenditure to reduce the long-term operational costs. From the optimization, the ice rink will ensure better financial sustainability in the long-term expectations.

Environmental Footprint

The system will use direct and indirect greenhouse gas emissions to find the environmental impact of the refrigeration cycle used. There may be direct emissions caused by refrigerant leaks. Using R-513A for the refrigerant instead of refrigerants like R-22, the Global Warming Potential of the system is reduced [16]. Using a 30% propylene glycol secondary loop, the primary refrigerant is isolated to the mechanical room. The ice rink system designed will prevent leaks into the arena, to prevent safety and financial hazards.

The system's electrical consumption determines the indirect emissions. Since Missouri's regional power grid still utilizes fossil fuel generation, it is important to maximize the system's COP [17]. Maximizing the efficiency directly reduces the required compressor power and lowers the carbon footprint.

System Enhancement: Waste Heat Recovery

In order to create a more optimal ice rink facility in terms of financial and environmental impact, the facility will incorporate a heat recovery desuperheater [18].

The condenser will be air-cooled and specifically designed to reject heat at 41.1°C (106°F). A heat exchanger will be used at the compressor discharge in order to capture this higher-grade thermal energy, rather than exhausting it into the 32.8°C outdoor summer air. The recovered energy will be used as a pre-heater for the ice rink facility's hot water supply. This heated water is necessary for resurfacing the ice and helping maintain comfortable air temperatures. By using this system enhancement, the ice rink facility's reliance on natural gas heating will decrease significantly, which lowers the overall operational expenditures and shrinks the facility's carbon footprint.

7 Conclusion and Sanity Check

Final answers are provided in Table 1:

$\dot{Q}_{cool}$ (kW)	379
COP	2.82
L_{evap} (m)	15.2

Table 1: Final results

These parameters were compared against standard real-world ice rink engineering guidelines to sanity check calculations:

- **Cooling Load (Tons of Refrigeration):** The calculated peak cooling requirement of 379 kW converts to approximately 90 Tons of Refrigeration (TR) (1 TR = 3.517 kW). According to the Ontario Recreation Facilities Association (ORFA), a standard 85' $\times$ 200' regulation ice surface requires a minimum baseline of 85 TR [19]. Our calculated value of 90 TR falls perfectly within this standard operating window, ensuring sufficient capacity for peak St. Louis summer conditions.
- **Coefficient of Performance (COP):** Baseline secondary loop systems often operate with COPs ranging from 2.0 to 2.5, while highly optimized or direct-refrigeration plants can push slightly past 3.0 [20]. A COP of 2.82 indicates that our selected evaporator pressures successfully minimized compressor work, placing our design strictly above average efficiency standards for indirect rinks.
- **Length of exchanger** When compared to typical shell and tube heat exchangers, this length is slightly larger than normal (15m compared to a high end of 6m) [21]. However, it is within the same order of magnitude, and likely based on the large temperature decrease in the brine.

In conclusion, by optimizing the pressure and utilizing waste heat recovery, the proposed R-513A/glycol chiller design meets all necessary safety and thermal requirements. The resulting system presents a highly realistic, environmentally conscious, and financially viable solution for year-round ice rink operation in the Midwest.

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A Appendix: Code 1

1. Calculation of $\dot{Q}_{cool}$

```
clear; clc;

% Geometry
L = 61;
W = 26;
A_ice = L * W;

% Temperatures (K)
T_air = 273.15 + 12.8;
T_ice = 273.15 - 5.5;
T_sur = 273.15 + 15;
T_f = 273.15 + 40;

% Convection
V_air = 0.2;
alpha_c = 3.41 + 3.55 * V_air;
Q_conv = alpha_c * A_ice * (T_air - T_ice)

% Diffusion
phi = 0.5;
T_air_C = T_air - 273.15;
T_ice_C = T_ice - 273.15;
P1 = 0.61078 * exp((17.27 * T_air_C) / (T_air_C + 237.3)) /
    100;
P2 = 0.61078 * exp((17.27 * T_ice_C) / (T_ice_C + 237.3)) /
    100;
DeltaP = phi * P1 - P2;
DeltaT = T_air - T_ice;
alpha_d = 1740 * alpha_c * (DeltaP / DeltaT);
Q_diffusion = alpha_d * A_ice * DeltaT

% Radiation
epsilon = 0.95;
```



```

sigma = 5.6704e-8;
Q_rad = epsilon * sigma * A_ice * (T_sur^4 - T_ice^4)

% Resurfacing
V_f = 0.5;
T_f_C = T_f - 273.15;
T_ice_C = T_ice - 273.15;
Q_resurface = 1000 * V_f * (4.2 * (T_f_C - 0) + 334 + 2 *
    (0 - T_ice_C));
t_freeze = 3600;
Q_resurface = Q_resurface * 1000 / t_freeze

% Total cooling load
Q_cool = Q_conv + Q_rad + Q_diffusion + Q_resurface
U_total = Q_cool / (A_ice * (T_air - T_ice))

```

The below Python script was utilized to solve for the length of the shell and tube heat exchanger. It was created with the use of AI [22].

```

import math
import numpy as np
from CoolProp.CoolProp import PropsSI

def NTU_from_eps(eps):
    if eps >= 1.0:
        print("This is the eps we are trying for: " + str(eps))
        raise ValueError("epsilon >= 1 -> requires infinite NTU
            (infinite area). Check temperatures/load.")
    return -np.log(1.0 - eps)

q_required = 379000.0    # W

# Brine temps [C]
T_brine_in_C = -5.5
T_brine_out_C = -11.1
T_ref_sat_C = -16.7    # R134a evaporating temperature

```

```

# Convert to K
T_bi  = T_brine_in_C  + 273.15
T_bo  = T_brine_out_C + 273.15
T_ref = T_ref_sat_C   + 273.15

P_brine = 1e5
P_ref   = 155100.0

# Selected Geometry
shellDiameter = 0.7552
D_inner       = 0.019
Sd            = 0.0238
Sm            = 0.0556
N_tubes       = 1000

T_brine_mean = 0.5 * (T_bi + T_bo)
fluid_brine  = "INCOMP::MPG-30%"

# Brine Property Evaluation
rho_b = PropsSI("D", "T", T_brine_mean, "P", P_brine,
               fluid_brine)
mu_b  = PropsSI("V", "T", T_brine_mean, "P", P_brine,
               fluid_brine)
k_b   = PropsSI("L", "T", T_brine_mean, "P", P_brine,
               fluid_brine)
cp_b  = PropsSI("C", "T", T_brine_mean, "P", P_brine,
               fluid_brine)

DeltaT_brine = abs(T_bo - T_bi)
m_dot_brine  = q_required / (cp_b * DeltaT_brine)

A_tube = math.pi * (D_inner**2) / 4.0
V_b     = (m_dot_brine / N_tubes) / (rho_b * A_tube)

# Tube-side heat transfer
Re_b = rho_b * V_b * D_inner / mu_b

```

```

Pr_b = cp_b * mu_b / k_b
Nu_b = 3.66
h_tube = Nu_b * k_b / D_inner

# Shell-side heat transfer (R134a)
fluid_ref = "R134a"
rho_ref = (PropsSI("D", "Q", 0.38, "P", P_ref, fluid_ref) +
            PropsSI("D", "Q", 1.0, "P", P_ref, fluid_ref)) / 2
mu_ref = (PropsSI("V", "Q", 0.38, "P", P_ref, fluid_ref) +
            PropsSI("V", "Q", 1.0, "P", P_ref, fluid_ref)) / 2
k_ref = (PropsSI("L", "Q", 0.38, "P", P_ref, fluid_ref) +
            PropsSI("L", "Q", 1.0, "P", P_ref, fluid_ref)) / 2

m_dot_ref = 2.147 # kg/s
Gs = m_dot_ref / Sm
De = 4.0 * (Sd**2 - math.pi * D_inner**2 / 4.0) / (math.pi *
            D_inner)

Re_shell = De * Gs / mu_ref
Pr_shell = cp_b * mu_ref / k_ref

Nu_shell = 0.36 * (Re_shell**0.55) * (Pr_shell**(1.0/3.0))
h_shell = Nu_shell * k_ref / De

# Overall Heat Transfer Coefficient
U = 1.0 / (1.0/h_tube + 1.0/h_shell)

# epsilon-NTU Calculation
C_brine = m_dot_brine * cp_b
C_min = C_brine
T_h_in = T_bi
T_c_in = T_ref

eps_target = q_required / (C_min * (T_h_in - T_c_in))
NTU_required = NTU_from_eps(eps_target)

```

```
L_required = NTU_required * C_min / (U * math.pi * D_inner *
    N_tubes)
```

```
# Output
```

```
print(f"epsilon_target_{uuuu}={eps_target:.4f}")
print(f"NTU_required_{uuuuuu}={NTU_required:.4f}")
print(f"L_required_{uuuuuuuu}={L_required:.3f}_m_per_tube")
```

The below MATLAB script was used to find and plot the system's efficiency against varying evaporating temperatures and pressures. The coding was created with the assistance of AI [23].

```
clear; clc;

% Our Inputs
h4 = 158.17;
T_brine_out = -11.1;

% Testing range from -25 C to -11.5 C for approach to brine supply temps
T1_range = -25:0.1:-11.5;
P_range = zeros(size(T1_range));
COP_results = zeros(size(T1_range));

for i = 1:length(T1_range)
    T1 = T1_range(i);

    % temp. to pressure in bar for R-134a
    P_range(i) = 0.0028 * T1^2 + 0.155 * T1 + 3.32;

    % COP results with linear approx. for enthalpy based on -16.7C
    % values that were hand calculated
    h1 = 288.52 + 0.62 * (T1 - (-16.7));
    ideal_work = 36.96 - 0.55 * (T1 - (-16.7));
    actual_work = ideal_work / 0.80;

    COP_results(i) = (h1 - h4) / actual_work;
end
```

```

% Plot
figure(1)

% Temperature vs COP Plot
subplot(2,1,1)
plot(T1_range, COP_results, 'LineWidth', 2); grid on;
hold on;
plot(-16.7, 2.82, 'ro', 'MarkerSize', 8, 'MarkerFaceColor', 'r'); % Your Point
ylabel('System COP');
xlabel('Evaporator Temperature T_1 (^{\circ}C)');
title('Thermodynamic Optimization: Efficiency vs. Saturation Conditions');
legend('Efficiency Curve', 'Selected Design (-16.7 ^{\circ}C)', 'Location', 'northwest');

% Pressure vs COP Plot
subplot(2,1,2)
plot(P_range, COP_results, 'LineWidth', 2); grid on;
hold on;
plot(1.50, 2.82, 'ro', 'MarkerSize', 8, 'MarkerFaceColor', 'r'); % Your Point
ylabel('System COP');
xlabel('Evaporator Pressure P_{low} (bar)');
legend('Efficiency Curve', 'Selected Design (1.50 bar)', 'Location', 'northwest');

```

B Appendix: Calculations

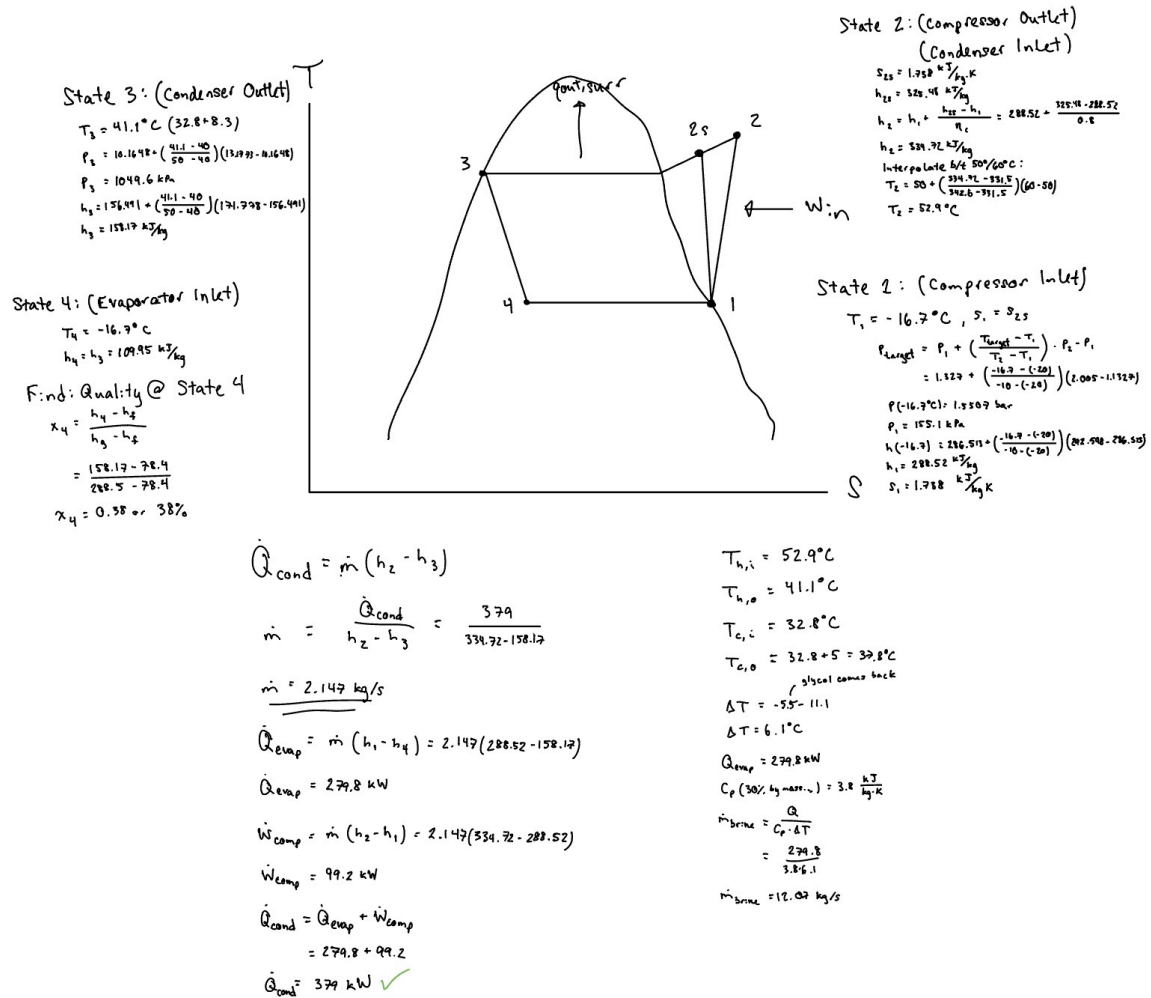


Figure 3: Thermodynamic Calculations and Properties for the States of the Ice Rink System.

C Appendix: Equations

Equation	Variables (Name & Units)
$\dot{Q}_{cool} = \dot{Q}_{conv} + \dot{Q}_{rad} + \dot{Q}_{diff} + \dot{Q}_{resurface}$	$\dot{Q}$: heat transfer rate (W)
$\dot{Q}_{conv} = h_{conv}A_{ice}(T_{air} - T_{ice})$	h_{conv} : convection coefficient (W/m ² K), A_{ice} : area (m ²), T : temperature (K or °C)
$\dot{Q}_{rad} = \epsilon\sigma A_{ice}(T_{ceiling}^4 - T_{ice}^4)$	ϵ : emissivity, σ : Stefan-Boltzmann constant (W/m ² K ⁴), A : area (m ²), T : temperature (K)
$\dot{Q}_{diff} = h_{diff}A_{ice}(T_{air} - T_{ice})$	h_{diff} : diffusion coefficient (W/m ² K), A : area (m ²), T : temperature (K or °C)
$h_{diff} = 1740(3.41 + 3.55v_{air}) \left(\frac{\phi P_{air} - P_{ice}}{T_{air} - T_{ice}} \right)$	v_{air} : air velocity (m/s), ϕ : relative humidity, P : vapor pressure (Pa), T : temperature (K or °C)
$\dot{Q}_{resurface} = 1000V_f[4.2(T_{water}) + 334 - 2T_{ice}]$	V_f : water volume (m ³), T : temperature (°C), constants in kJ/kg
$\dot{Q} = \dot{m}(h_{out} - h_{in})$	$\dot{m}$: mass flow rate (kg/s), h : specific enthalpy (kJ/kg)
$\Delta T_{lm} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1 / \Delta T_2)}$	ΔT : temperature difference (K)
$\Delta T = \frac{\dot{Q}}{\dot{m}c_p}$	c_p : specific heat (kJ/kgK), $\dot{m}$: mass flow (kg/s)
$L = \frac{A}{\pi DN}$	L : length (m), D : diameter (m), N : number of tubes, A : area (m ²)
$D_h = D_{outer} - D_{inner}$	D : diameter (m)
$S_m = (\text{baffle spacing})(S_D - D)$	S_m : crossflow area (m ²), S_D : pitch spacing (m), D : diameter (m)
$V = \frac{\dot{m}}{\rho A}$	V : velocity (m/s), ρ : density (kg/m ³), A : area (m ²)
$Re = \frac{\rho VD}{\mu}$	Re : Reynolds number, μ : viscosity (Pa·s)
$Pr = \frac{c_p \mu}{k}$	Pr : Prandtl number, k : thermal conductivity (W/mK)
$\bar{Nu} = 3.66$	Nu : Nusselt number
$Nu = \frac{hD}{k}$	h : convection coeff. (W/m ² K), D : diameter (m), k : conductivity (W/mK)
$\Delta P = f \frac{L}{D} \frac{\rho u^2}{2}$	ΔP : pressure drop (Pa), f : friction factor, u : velocity (m/s)
$A = \frac{\pi D^2}{4}$	A : area (m ²), D : diameter (m)
$\dot{m}_{cool} = \frac{\dot{Q}_{cond}}{c_p(T_{out} - T_{in})}$	$\dot{m}_{cool}$: coolant flow (kg/s), c_p : specific heat (kJ/kgK)
$\dot{W}_{cond} = \frac{\dot{m}_{cool}\Delta P}{\rho}$	$\dot{W}$: power (W), ΔP : pressure (Pa), ρ : density (kg/m ³)
$COP = \frac{\dot{Q}_{evap}}{\dot{W}_{comp}}$	COP : coefficient of performance, $\dot{W}$: power (W)
$q_{max} = C_{min}(T_{h,i} - T_{c,i})$	C_{min} : heat capacity rate (W/K), T : temperature (K)
$\epsilon = \frac{q}{q_{max}}$	ϵ : effectiveness, q : heat transfer (W)
$\epsilon = 1 - e^{-NTU}$	NTU : number of transfer units

Table 2: Summary of Governing Equations and Variables